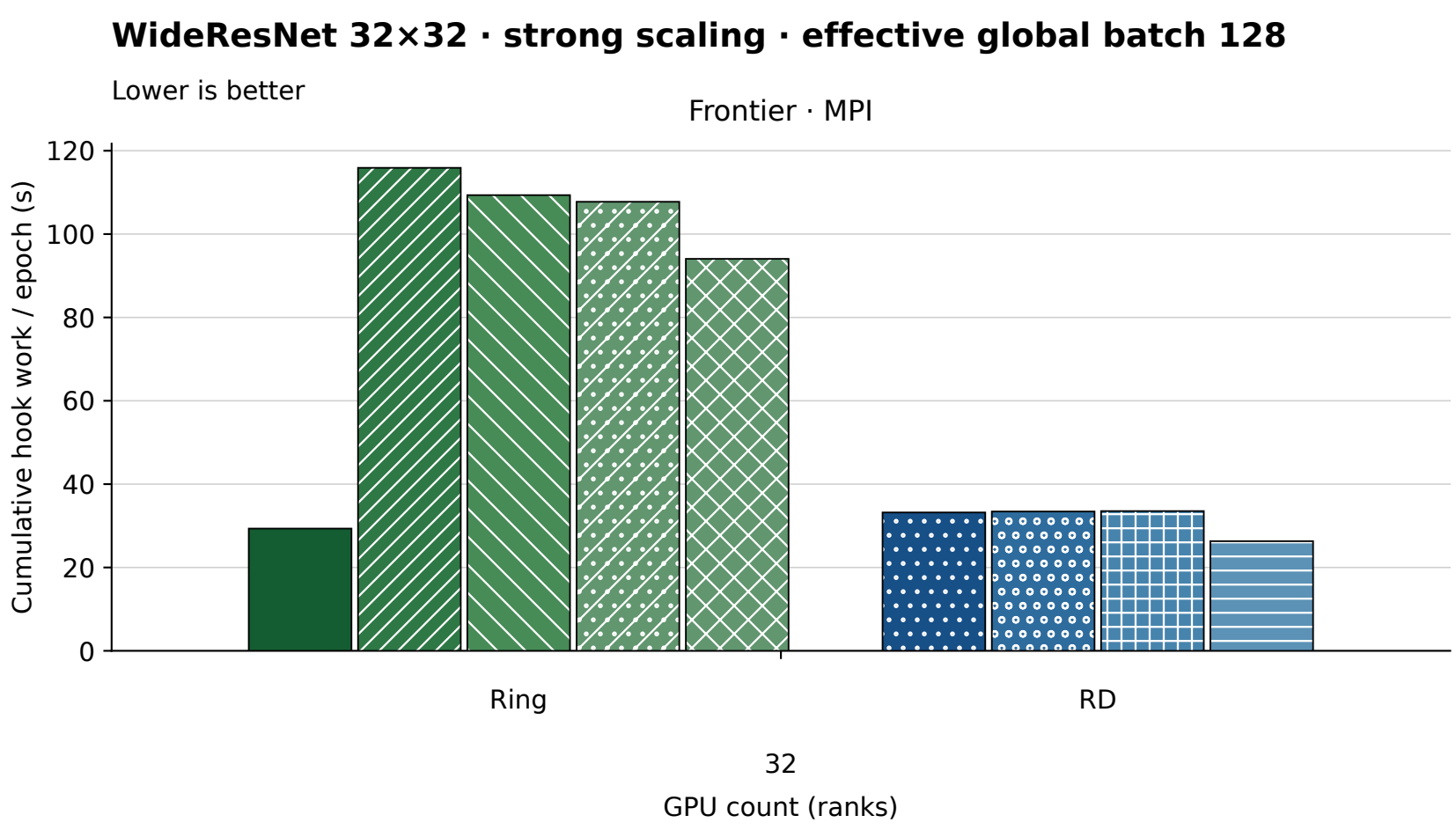

WideResNet 32×32 · strong scaling · effective global batch 128
Lower is better
Frontier · MPI
Cumulative hook work / epoch (s)
120
100
80
60
40
20
0
Ring
RD
32
GPU count (ranks)
Ring
Uncompressed
ZFP naive · rate16
ZFP online · rate16
ZFP online · rate10
ZFP online · rate8
Recursive doubling
Uncompressed
ZFP naive · rate16
ZFP online · rate16
ZFP online · rate8
Epoch 1 excluded · missing metrics are not zero · r = ZFP rate